

Limits (L_1)

The Limit of a Function

To find the tangent to a curve or the velocity of an object, we now turn our attention to limits in general and numerical and graphical methods for computing them.

Let's investigate the behavior of the function f defined by $f(x) = x^2 - x + 2$ for values of x near 2.

The Limit of a Function

The following table gives values of $f(x)$ for values of x close to 2 but not equal to 2.

x	$f(x)$	x	$f(x)$
1.0	2.000000	3.0	8.000000
1.5	2.750000	2.5	5.750000
1.8	3.440000	2.2	4.640000
1.9	3.710000	2.1	4.310000
1.95	3.852500	2.05	4.152500
1.99	3.970100	2.01	4.030100
1.995	3.985025	2.005	4.015025
1.999	3.997001	2.001	4.003001

The Limit of a Function

From the table and the graph of f (a parabola) shown in Figure 1 we see that when x is close to 2 (on either side of 2), $f(x)$ is close to 4.

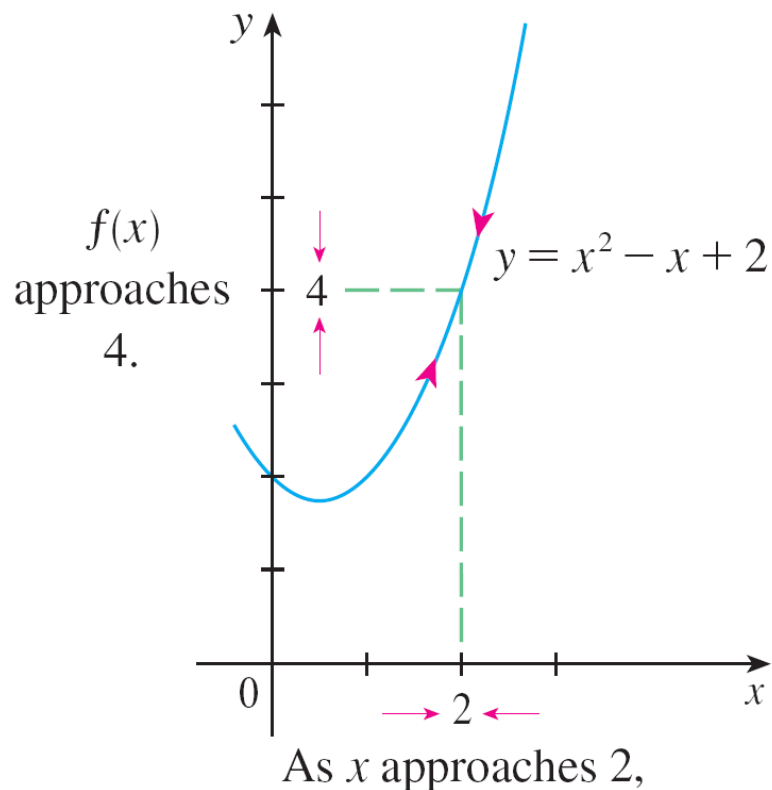


Figure 1

The Limit of a Function

In fact, it appears that we can make the values of $f(x)$ as close as we like to 4 by taking x sufficiently close to 2.

We express this by saying “the limit of the function $f(x) = x^2 - x + 2$ as x approaches 2 is equal to 4.”

The notation for this is

$$\lim_{x \rightarrow 2} (x^2 - x + 2) = 4$$

The Limit of a Function

In general, we use the following notation.

1 Definition We write

$$\lim_{x \rightarrow a} f(x) = L$$

and say “the limit of $f(x)$, as x approaches a , equals L ”

if we can make the values of $f(x)$ arbitrarily close to L (as close to L as we like) by taking x to be sufficiently close to a (on either side of a) but not equal to a .

This says that the values of $f(x)$ tend to get closer and closer to the number L as x gets closer and closer to the number a (from either side of a) but $x \neq a$.

The Limit of a Function

An alternative notation for

$$\lim_{x \rightarrow a} f(x) = L$$

is $f(x) \rightarrow L$ as $x \rightarrow a$

which is usually read “ $f(x)$ approaches L as x approaches a .”

Notice the phrase “but $x \neq a$ ” in the definition of limit. This means that in finding the limit of $f(x)$ as x approaches a , we never consider $x = a$. In fact, $f(x)$ need not even be defined when $x = a$. The only thing that matters is how f is defined *near* a .

Recall: Left-Hand Limit

We write

$$\lim_{x \rightarrow a^-} f(x) = L$$

and say “the **left-hand limit** of $f(x)$, as x approaches a , equals L ”

if we can make the values of $f(x)$ arbitrarily **close** to L (as close as we like) by taking x to be **sufficiently close** to a and x less than a .

Recall: Right-Hand Limit

We write

$\lim_{x \rightarrow a^+} f(x) = L$
and say “the **right-hand limit** of $f(x)$, as x approaches a , equals L ”

if we can make the values of $f(x)$ arbitrarily **close** to L (as close as we like) by taking x to be **sufficiently close** to a and x greater than a .

Recall: Limit of a Function

$$\lim_{x \rightarrow a} f(x) = L$$

if and only if

$$\lim_{x \rightarrow a^-} f(x) = L$$

Independent of $f(a)$

and

$$\lim_{x \rightarrow a^+} f(x) = L$$

Example 1

Guess the value of $\lim_{x \rightarrow 1} \frac{x - 1}{x^2 - 1}$.

Solution:

Notice that the function $f(x) = (x - 1)/(x^2 - 1)$ is not defined when $x = 1$, but that doesn't matter because the definition of $\lim_{x \rightarrow a} f(x)$ says that we consider values of x that are close to a but not equal to a .

Example 1 – *Solution*

The tables below give values of $f(x)$ (correct to six decimal places) for values of x that approach 1 (but are not equal to 1).

$x < 1$	$f(x)$
0.5	0.666667
0.9	0.526316
0.99	0.502513
0.999	0.500250
0.9999	0.500025

$x > 1$	$f(x)$
1.5	0.400000
1.1	0.476190
1.01	0.497512
1.001	0.499750
1.0001	0.499975

On the basis of the values in the tables, we make the guess that

$$\lim_{x \rightarrow 1} \frac{x - 1}{x^2 - 1} = 0.5$$

The Limit of a Function

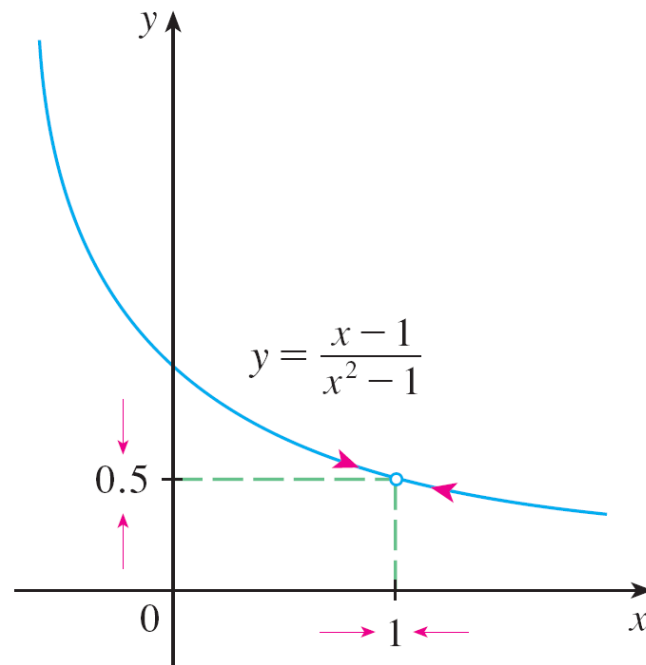


Figure 3

One-Sided Limits

One-Sided Limits

The function H is defined by

$$H(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$

$H(t)$ approaches 0 as t approaches 0 from the left and $H(t)$ approaches 1 as t approaches 0 from the right.

We indicate this situation symbolically by writing

$$\lim_{t \rightarrow 0^-} H(t) = 0 \quad \text{and} \quad \lim_{t \rightarrow 0^+} H(t) = 1$$

One-Sided Limits

The symbol “ $t \rightarrow 0^-$ ” indicates that we consider only values of t that are less than 0.

Likewise, “ $t \rightarrow 0^+$ ” indicates that we consider only values of t that are greater than 0.

One-Sided Limits

2 **Definition** We write

$$\lim_{x \rightarrow a^-} f(x) = L$$

and say the **left-hand limit of $f(x)$ as x approaches a** [or the **limit of $f(x)$ as x approaches a from the left**] is equal to L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a and x less than a .

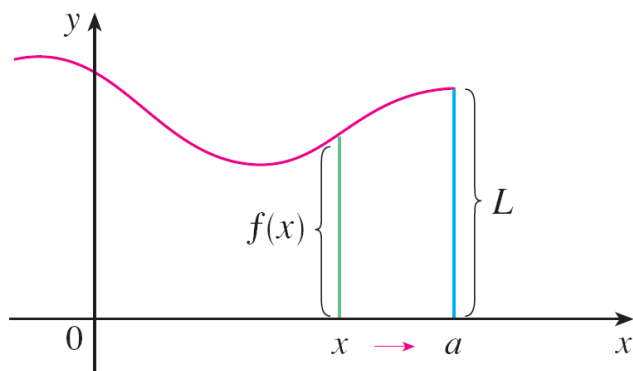
Notice that Definition 2 differs from Definition 1 only in that we require x to be less than a .

One-Sided Limits

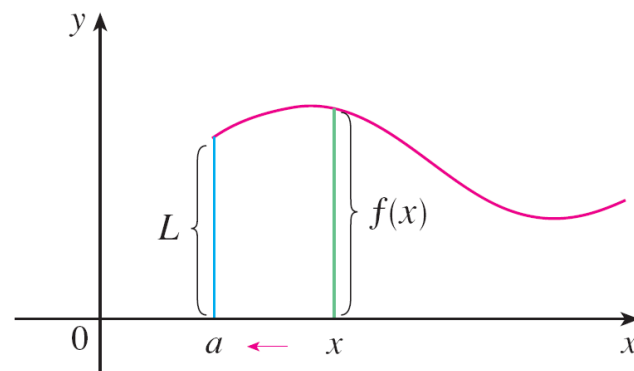
Similarly, if we require that x be greater than a , we get “the **right-hand limit of $f(x)$ as x approaches a** is equal to L ” and we write

$$\lim_{x \rightarrow a^+} f(x) = L$$

Thus the symbol “ $x \rightarrow a^+$ ” means that we consider only $x > a$. These definitions are illustrated in Figure 9.



(a) $\lim_{x \rightarrow a^-} f(x) = L$



(b) $\lim_{x \rightarrow a^+} f(x) = L$

Figure 9

One-Sided Limits

By comparing Definition 1 with the definitions of one-sided limits, we see that the following is true.

$$\boxed{3} \quad \lim_{x \rightarrow a} f(x) = L \quad \text{if and only if} \quad \lim_{x \rightarrow a^-} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = L$$

Example

Find

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

Explanation: $f(x) = \frac{x^2 - 4}{x - 2}$ and $a = 2$ from the table above we have the following

x	$f(x)$		x	$f(x)$
$2 - 0.1 = 1.9$	$f(2 - 0.1) = f(1.9) = 3.9$		$2 + 0.1 = 2.1$	$f(2 + 0.1) = f(2.1) = 4.1$
$2 - 0.01 = 1.99$	$f(2 - 0.01) = f(1.99) = 3.99$		$2 + 0.01 = 2.01$	$f(2 + 0.01) = f(2.01) = 4.01$
$2 - 0.001 = 1.999$	$f(2 - 0.001) = f(1.999) = 3.999$		$2 + 0.001 = 2.001$	$f(2 + 0.001) = f(2.001) = 4.001$
$2 - 0.0001 = 1.9999$	$f(2 - 0.0001) = f(1.9999) = 3.9999$		$2 + 0.0001 = 2.0001$	$f(2 + 0.0001) = f(2.0001) = 4.0001$

Class Activity:

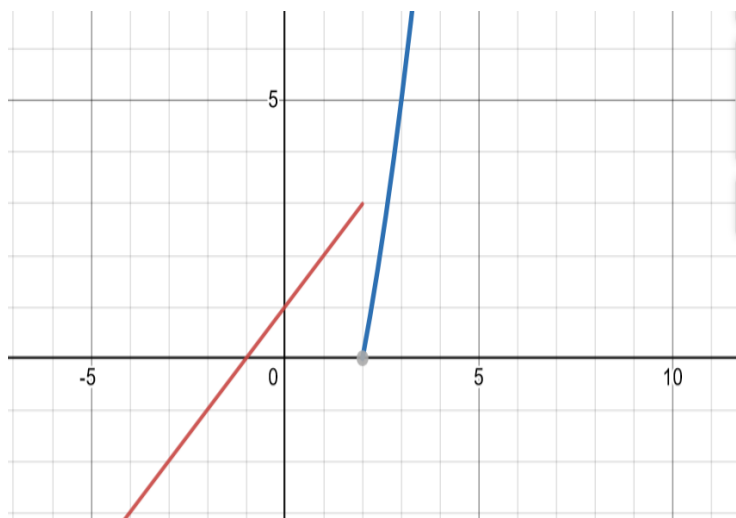
Find the limit using the graph of functions.

Evaluating One-Sided Limits

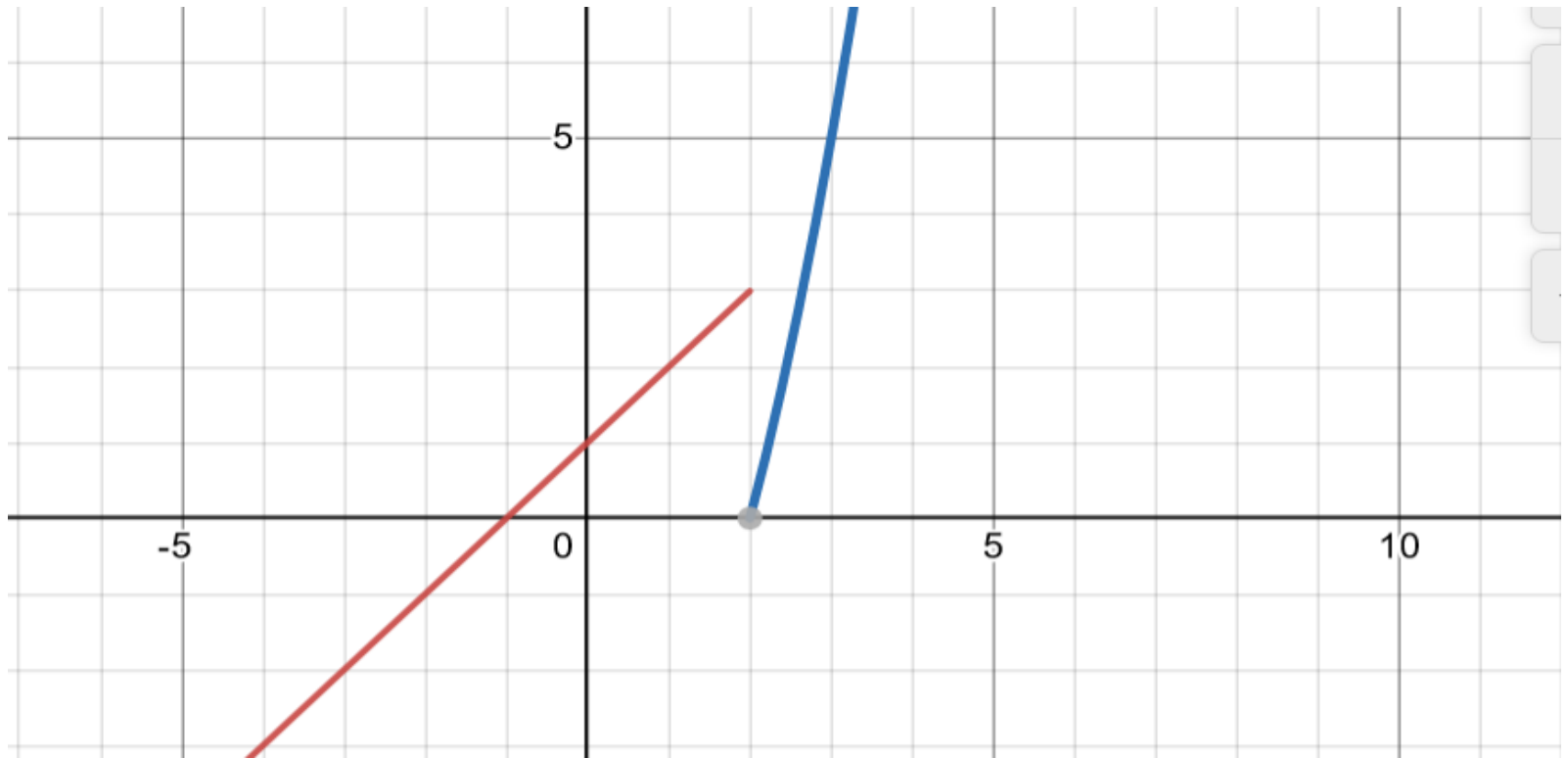
For the function $f(x) = \begin{cases} x + 1 & \text{if } x < 2 \\ x^2 - 4 & \text{if } x \geq 2 \end{cases}$, evaluate each of the following limits.

a. $\lim_{x \rightarrow 2^-} f(x)$

b. $\lim_{x \rightarrow 2^+} f(x)$



Graphically:



Using Table:

$$f(x) = x + 1$$

$$f(x) = x^2 - 4$$

x	$f(x) = x + 1$		x	$f(x) = x^2 - 4$
1.9	2.9		2.1	0.41
1.99	2.99		2.01	0.0401
1.999	2.999		2.001	0.004001
1.9999	2.9999		2.0001	0.00040001
1.99999	2.99999		2.00001	0.0000400001

Explanation:

$\lim_{x \rightarrow 2^-} f(x) = 3$ (From the table, we noted that x gets closer to 2 from the left then the values of f approaches 3)

$\lim_{x \rightarrow 2^+} f(x) = 0$ (From the table, we noted that x gets closer to 2 from the right then the values of f approaches 0)

$$\lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$$

The limit from the left does not equal the limit from the

Summary of Limit Laws

Limit Law in symbols	Limit Law in words
$\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$	The limit of a sum is equal to the sum of the limits.
$\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$	The limit of a difference is equal to the difference of the limits.
$\lim_{x \rightarrow a} cf(x) = c \lim_{x \rightarrow a} f(x)$	The limit of a constant times a function is equal to the constant times the limit of the function.
$\lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$	The limit of a product is equal to the product of the limits.
$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \left(\text{if } \lim_{x \rightarrow a} g(x) \neq 0 \right)$	The limit of a quotient is equal to the quotient of the limits.

Example

Use the limit laws to evaluate

$$\lim_{x \rightarrow 2} \frac{2x^2 - 3x + 1}{x^3 + 4}.$$

Solution

To find this limit, we need to apply the limit laws several times. Again, we need to keep in mind that as we rewrite the limit in terms of other limits, each new limit must exist for the limit law to be applied.

$$\lim_{x \rightarrow 2} \frac{2x^2 - 3x + 1}{x^3 + 4} = \frac{\lim_{x \rightarrow 2} (2x^2 - 3x + 1)}{\lim_{x \rightarrow 2} (x^3 + 4)} \quad \text{Apply the quotient law, make sure that } (2)^3 + 4 \neq 0.$$

Example cont.

$$= \frac{2 \cdot \lim_{x \rightarrow 2} x^2 - 3 \cdot \lim_{x \rightarrow 2} x + \lim_{x \rightarrow 2} 1}{\lim_{x \rightarrow 2} x^3 + \lim_{x \rightarrow 2} 4} \quad \text{Apply the sum law and constant multiple law}$$

$$= \frac{2 \cdot (\lim_{x \rightarrow 2} x)^2 - 3 \cdot \lim_{x \rightarrow 2} x + \lim_{x \rightarrow 2} 1}{(\lim_{x \rightarrow 2} x)^3 + \lim_{x \rightarrow 2} 4} \quad \text{Apply the power law.}$$

$$= \frac{2(4) - 3(2) + 1}{(2)^3 + 4} = \frac{1}{4}. \quad \text{Apply the basic limit laws and simplify.}$$

Notice this is equivalent to substituting 2 for x in the original function. One just needs to be careful that the limit exists at this point.

Example

Use the limit laws to evaluate $\lim_{x \rightarrow 6} (2x - 1) \sqrt{x + 4}$. In each step, indicate the limit law applied.

Solution:

$\lim_{x \rightarrow 6} (2x - 1) \sqrt{x + 4} = \lim_{x \rightarrow 6} (2x - 1) \cdot \lim_{x \rightarrow 6} \sqrt{x + 4}$. Apply Product law for limits,

$= (2 \cdot \lim_{x \rightarrow 6} x - \lim_{x \rightarrow 6} 1) \cdot \lim_{x \rightarrow 6} \sqrt{x + 4}$. Apply the constant

multiple law

$$= (2(6) - 1) \cdot \sqrt{6 + 4}$$

$$= 11\sqrt{10}$$

Example

Evaluating a Limit of a Rational Function

Evaluate the $\lim_{x \rightarrow 3} \frac{2x^2 - 3x + 1}{5x + 4}$.

Solution

Since 3 is in the domain of the rational function $f(x) = \frac{2x^2 - 3x + 1}{5x + 4}$, we can calculate the limit by substituting 3 for x into the function. Thus,

$$\lim_{x \rightarrow 3} \frac{2x^2 - 3x + 1}{5x + 4} = \frac{10}{19}.$$

Evaluating a Limit by Factoring and Canceling

Example

Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 3x}{2x^2 - 5x - 3}$.

Solution

Step 1. The function $f(x) = \frac{x^2 - 3x}{2x^2 - 5x - 3}$ is undefined for $x = 3$. In fact, if we substitute 3 into the function we get $0/0$, which is undefined. Factoring and canceling is a good strategy:

Example cont.

$$\lim_{x \rightarrow 3} \frac{x^2 - 3x}{2x^2 - 5x - 3} = \lim_{x \rightarrow 3} \frac{x(x - 3)}{(x - 3)(2x + 1)}$$

Step 2. For all $x \neq 3$, $\frac{x^2 - 3x}{2x^2 - 5x - 3} = \frac{x}{2x + 1}$. Therefore,

$$\lim_{x \rightarrow 3} \frac{x(x - 3)}{(x - 3)(2x + 1)} = \lim_{x \rightarrow 3} \frac{x}{2x + 1}.$$

Step 3. Evaluate using the limit laws:

$$\lim_{x \rightarrow 3} \frac{x}{2x + 1} = \frac{3}{7}.$$

Evaluating a Limit by Multiplying by a conjugate

Evaluate $\lim_{x \rightarrow -1} \frac{\sqrt{x+2}-1}{x+1}$.

Solution

Step 1. $\frac{\sqrt{x+2}-1}{x+1}$ has the form $0/0$ at -1 . Let's begin by multiplying by $\sqrt{x+2}+1$, the conjugate of $\sqrt{x+2}-1$, on the numerator and denominator:

$$\lim_{x \rightarrow -1} \frac{\sqrt{x+2}-1}{x+1} = \lim_{x \rightarrow -1} \frac{\sqrt{x+2}-1}{x+1} \cdot \frac{\sqrt{x+2}+1}{\sqrt{x+2}+1}.$$

Step 2. We then multiply out the numerator. We don't multiply out the denominator because we are hoping that the $(x+1)$ in the denominator cancels out in the end:

$$= \lim_{x \rightarrow -1} \frac{x+1}{(x+1)(\sqrt{x+2}+1)}.$$

Evaluating a Limit when the Limit Laws Do Not Apply

Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{x} + \frac{5}{x(x-5)} \right)$.

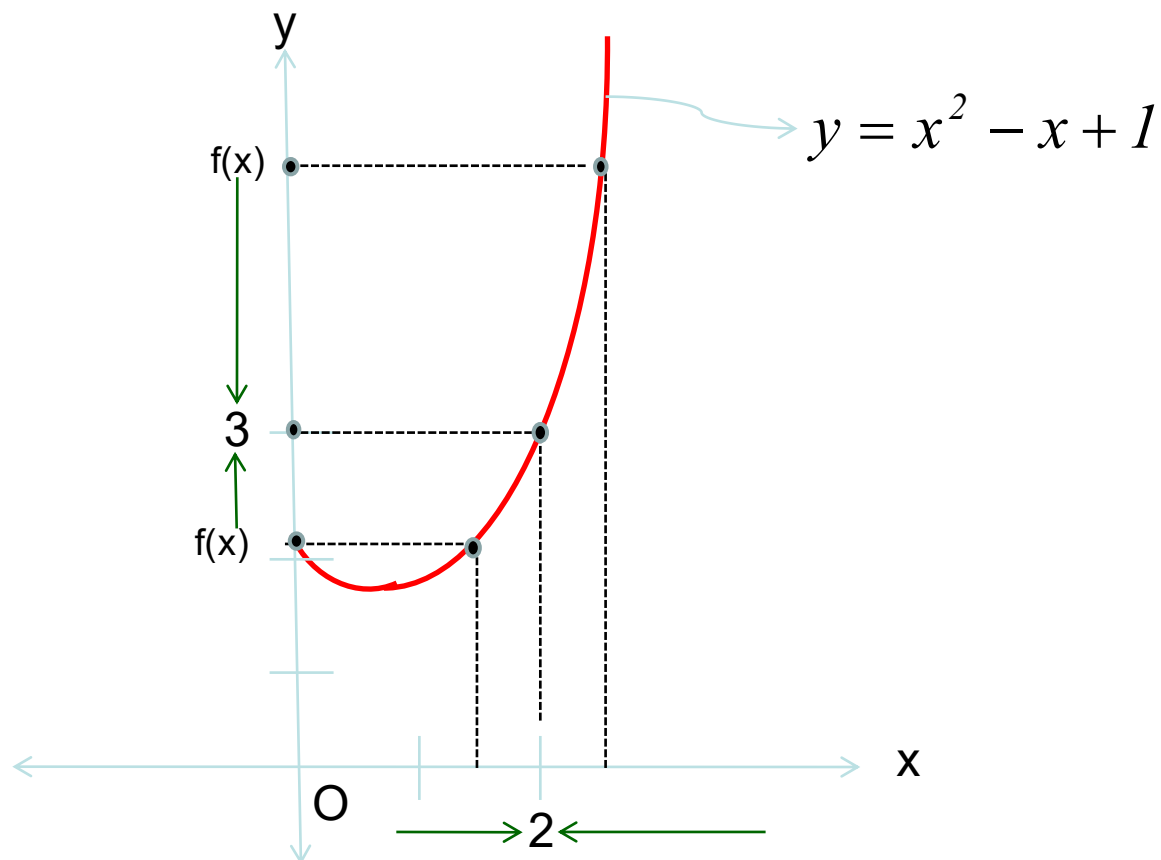
Solution

Both $1/x$ and $5/x(x-5)$ fail to have a limit at zero. Since neither of the two functions has a limit at zero, we cannot apply the sum law for limits; we must use a different strategy. In this case, we find the limit by performing addition and then applying one of our previous strategies. Observe that

$$\begin{aligned} \frac{1}{x} + \frac{5}{x(x-5)} &= \frac{x-5+5}{x(x-5)} \\ &= \frac{x}{x(x-5)}. \end{aligned}$$

Example cont.

$$\begin{aligned}\lim_{x \rightarrow 0} \left(\frac{1}{x} + \frac{5}{x(x-5)} \right) &= \lim_{x \rightarrow 0} \frac{x}{x(x-5)} \\ &= \lim_{x \rightarrow 0} \frac{1}{x-5} \\ &= -\frac{1}{5}.\end{aligned}$$



x	1.9	1.95	1.99	1.995	1.999	2	2.001	2.005	2.01	2.05	2.1
F(x)	2.71	2.85	2.97	2.98	2.99		3.00	3.01	3.03	3.15	3.31
		2	left side				right side				

Basic tool for finding limits algebraically.

1.2.2 THEOREM *Let a be a real number, and suppose that*

$$\lim_{x \rightarrow a} f(x) = L_1 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = L_2$$

That is, the limits exist and have values L_1 and L_2 , respectively. Then:

(a) $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = L_1 + L_2$

(b) $\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) = L_1 - L_2$

(c) $\lim_{x \rightarrow a} [f(x)g(x)] = \left(\lim_{x \rightarrow a} f(x) \right) \left(\lim_{x \rightarrow a} g(x) \right) = L_1 L_2$

(d) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L_1}{L_2}, \quad \text{provided } L_2 \neq 0$

(e) $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} = \sqrt[n]{L_1}, \quad \text{provided } L_1 > 0 \text{ if } n \text{ is even.}$

Moreover, these statements are also true for the one-sided limits as $x \rightarrow a^-$ or as $x \rightarrow a^+$.

$$2. \lim_{x \rightarrow 0} \frac{\sqrt{x+2} - \sqrt{2}}{x}$$

Solution:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+2} - \sqrt{2}}{x} = \frac{\sqrt{0+2} - \sqrt{2}}{0} = \frac{0}{0} \text{ (indeterminate)}$$

Rationalizing the numerator:

$$= \lim_{x \rightarrow 0} \frac{\sqrt{x+2} - \sqrt{2}}{x} \cdot \frac{\sqrt{x+2} + \sqrt{2}}{\sqrt{x+2} + \sqrt{2}} = \lim_{x \rightarrow 0} \frac{x+2-2}{x(\sqrt{x+2} + \sqrt{2})}$$

$$= \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x+2} + \sqrt{2})} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+2} + \sqrt{2}} = \frac{1}{\sqrt{2} + \sqrt{2}} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}$$

$$\therefore \lim_{x \rightarrow 0} \frac{\sqrt{x+2} - \sqrt{2}}{x} = \frac{\sqrt{2}}{4}$$

$$3. \lim_{x \rightarrow \frac{3}{2}} \sqrt{\frac{8x^3 - 27}{4x^2 - 9}}$$

Solution:

$$\lim_{x \rightarrow \frac{3}{2}} \sqrt{\frac{8x^3 - 27}{4x^2 - 9}} = \sqrt{\frac{8\left(\frac{3}{2}\right)^3 - 27}{4\left(\frac{3}{2}\right)^2 - 9}} = \sqrt{\frac{27 - 27}{9 - 9}} = \frac{0}{0} \quad \text{(indeterminate)}$$

By Factoring:

$$\begin{aligned} &= \lim_{x \rightarrow \frac{3}{2}} \sqrt{\frac{(2x-3)(4x^2+6x+9)}{(2x+3)(2x-3)}} = \lim_{x \rightarrow \frac{3}{2}} \sqrt{\frac{4x^2+6x+9}{(2x+3)}} = \sqrt{\frac{4\left(\frac{3}{2}\right)^2 + 6\left(\frac{3}{2}\right) + 9}{\left(2\left(\frac{3}{2}\right) + 3\right)}} \\ &= \sqrt{\frac{9+9+9}{3+3}} = \sqrt{\frac{27}{6}} = \sqrt{\frac{9}{2}} = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2} \end{aligned}$$

$$\therefore \lim_{x \rightarrow \frac{3}{2}} \sqrt{\frac{8x^3 - 27}{4x^2 - 9}} = \frac{3\sqrt{2}}{2}$$